Táboa de Integrais Inmediatas

Función	*Primitiva*	Función	*Primitiva*
$[f(x)]^m\ f'(x),\ m\neq-1$	$\frac{1}{m+1}\ [f(x)]^{m+1}$	$\frac{f'(x)}{f(x)}$	$ln\lvert f(x)\rvert$
$a^{f(x)}\ f'(x)$	$\frac{a^{f(x)}}{ln(a)}$	$sen[f(x)]f'(x)$	$-cos[f(x)]$
$cos[f(x)]f'(x)$	$sen[f(x)]$	$tg[f(x)]f'(x)$	$-ln\lvert cos[f(x)]\rvert$
$cotg[f(x)]f'(x)$	$ln\lvert sen[f(x)]\rvert$	$\frac{f'(x)}{cos^2[f(x)]}$	$tg[f(x)]$
$\frac{f'(x)}{sen^2[f(x)]}$	$-cotg[f(x)]$	$\frac{f'(x)}{\sqrt{1-[f(x)]^2}}$	$arcsen[f(x)]$
$\frac{f'(x)}{1+[f(x)]^2}$	$arctg[f(x)]$		

Integrais trigonométricas $\int R(sen(x), cos(x))\, dx$

Troco universal: $t=tg\left(\frac{x}{2}\right),\ dx=\frac{2}{1+t^2}\,dt\ ,\ sen(x)=\frac{2t}{1+t^2},\ cos(x)=\frac{1-t^2}{1+t^2}$

$\int R(sen(x))cos(x)dx\ ;\ t = sen(x)\ ,\ dt = cos(x)\ dx$

$\int R(cos(x))sen(x)\ dx\ ;\ t = cos(x)\ ,\ dt = -sen(x)\ dx$

$\int R(tg(x))\ dx\ ;\ t = tg(x)\ ,\ x = arctg(t),\ dt=\frac{dt}{1+t^2}$

$\int R(sen(x),cos(x))\ dx$ con $R(\ sen(\ x\ ),\ cos(\ x\ )\ ) = sen^m(x)\ cos^n(x)$

(a) se algún exponente é impar realízase o troco t = ***a outra función***

(b) se m, n son pares non negativos: $sen^2(x) = \frac{1-cos(2x)}{2}\ ,\ cos^2(x) = \frac{1+cos(2x)}{2}$

(c) se m, n son pares e algún é negativo:

$$t = tg(x),\ dx = \frac{1}{1+t^2}\,dt\ ,\ sen^2(x) = \frac{t^2}{1+t^2}\ ,\ cos^2(x) = \frac{1}{1+t^2}$$

$\int cos(px)\ cos(qx)\ dx\ ;\ \int sen(px)\ sen(qx)\ dx\ ;\ \int cos(px)\ sen(qx)\ dx$

$cos(px)\ cos(qx) = \frac{1}{2}\ [cos(p+q)x + cos(p-q)x]$

$sen(px)\ sen(qx) = \frac{1}{2}\ [-cos(p+q)x + cos(p-q)x]$

$sen(px)\ cos(qx) = \frac{1}{2}\ [sen(p+q)x + sen(p-q)x]$

Integrais racionais $\int \frac{P(x)}{Q(x)}\,dx$:

factorízase $Q(x)$ e escríbese $\frac{P(x)}{Q(x)}$ como suma de fraccións simples.

$$\int \frac{1}{x-a}\,dx = ln\ \lvert x-a\rvert$$

$$\int \frac{1}{(x-a)^p}\,dx = \frac{(x-a)^{-p+1}}{-p+1}$$

$$\int \frac{Mx+N}{(x-a)^2+b^2}\,dx = \frac{M}{2}\ ln\ \left|(x-a)^2+b^2\right| + \frac{N+Ma}{b}\ arctg\left(\frac{x-a}{b}\right)$$

Método de Hermite : $\int \frac{P(x)}{Q(x)}\,dx = \frac{Y(x)}{F(x)} + \int \frac{X(x)}{G(x)}\,dx$

sendo $F(x)$ = M.C.D.(Q,Q'), $G(x)=\frac{Q(x)}{F(x)}$

$$\int R\left(x,\left(\frac{ax+b}{cx+d}\right)^{\frac{m}{n}},\ldots,\left(\frac{ax+b}{cx+d}\right)^{\frac{r}{s}}\right)dx$$ troco $t^k=\dfrac{ax+b}{cx+d}$, sendo k = m.c.m.$(n,\ldots,s)$

$\int R\left(x,\sqrt{x^2+a^2}\right)dx$, $x=a\,tg(t)$, $\sqrt{x^2+a^2}=\dfrac{a}{\cos(t)}$, $dx=\dfrac{a}{cos^2(t)}\,dt$

$x=a\,cotg(t)$, $\sqrt{x^2+a^2}=\dfrac{a}{sen(t)}$, $dx=\dfrac{-a}{sen^2(t)}\,dt$

$\int R\left(x,\sqrt{x^2-a^2}\right)dx$, $x=a\,sec(t)$, $\sqrt{x^2-a^2}=a\,tg(t)$, $dx=a\,tg(t)\,sec(t)$

$x=a\,cosec(t)$, $\sqrt{x^2-a^2}=a\,cotg(t)$, $dx=-a\,cotg(t)\,cosec(t)\,dt$

$\int R\left(x,\sqrt{a^2-x^2}\right)dx$, $x=a\,sen(t)$, $\sqrt{a^2-x^2}=a\cos(t)$, $dx=a\cos(t)\,dt$

$x=a\cos(t)$, $\sqrt{a^2-x^2}=a\,sen(t)$, $dx=-a\,sen(t)\,dt$

$\int R\left(x,\sqrt{ax^2+bx+c}\right)dx$

$a>0$; $\sqrt{ax^2+bx+c}=\sqrt{a}\,x+t$, $x=\dfrac{t^2-c}{b-2\sqrt{a}\,t}$

$c>0$; $\sqrt{ax^2+bx+c}=x\,t+\sqrt{c}$, $x=\dfrac{2t\sqrt{c}-b}{a-t^2}$

se $a<0$ e r é unha raíz de ax^2+bx+c, o troco é $\sqrt{ax^2+bx+c}=(x-r)\,t$